

TORUNGEN FYR, Norway

WMO: 014650

Lat: 58.399N

Lon: 8.789E

Elev: 49

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.2	19.6	10.5	9.4	20.1	13.9	11.1	23.0	40.7	33.7	36.9	32.8	15.2	20	0.791

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	71.9	64.5	70.1	63.4	68.2	61.8	66.6	70.4	64.8	68.7	63.3	66.8	11.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	92.2	69.2	63.1	86.7	67.3	61.7	82.2	65.8	31.2	70.4	29.9	68.8	28.8	67.0	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.9	33.1	29.8	DB	14.2	75.6	5.1	3.3	10.5	78.0	7.5	79.9	4.7	81.7	0.9	84.1	(4)
(5)			WB	13.5	67.1	5.1	2.1	9.9	68.7	6.9	69.9	4.1	71.1	0.4	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	34.4	33.3	36.5	42.8	51.0	58.0	62.5	61.9	56.8	48.9	42.2	36.7	(6)	
	DBStd	11.68	6.59	6.65	5.78	4.32	4.93	3.98	3.58	3.22	3.64	5.11	5.96	7.77	(7)	
	HDD50	2374	484	467	420	221	45	1	0	0	2	82	239	413	(8)	
	HDD65	6550	949	887	885	667	436	212	97	108	246	500	685	878	(9)	
	CDD50	1327	0	0	0	3	74	241	386	368	205	46	4	0	(10)	
	CDD65	33	0	0	0	0	0	3	19	11	0	0	0	0	(11)	
	CDH74	16	0	0	0	0	0	5	9	2	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	15.0	17.1	15.5	15.3	13.6	13.8	13.4	13.0	13.3	14.9	16.3	17.0	16.7	(14)	
(15) Precipitation	PrecAvg	45.9	4.6	3.0	2.3	2.6	3.0	3.0	3.2	4.7	4.6	5.2	5.6	4.1	(15)	
	PrecMax	63.3	10.6	8.5	5.2	5.7	5.6	5.9	7.6	8.5	11.3	10.6	16.9	7.6	(16)	
	PrecMin	31.3	0.9	0.7	0.3	0.6	0.5	0.3	0.6	1.1	1.4	1.3	1.8	1.2	(17)	
	PrecStd	8.0	2.4	1.8	1.3	1.4	1.4	1.5	1.6	2.1	2.4	2.5	3.2	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.7	48.4	54.5	58.2	69.1	72.6	75.2	73.1	67.4	60.2	55.5	51.6	(19)	
		MCWB	46.3	43.0	46.1	48.7	63.0	61.2	66.3	64.9	58.6	55.5	52.8	49.3	(20)	
	2%	DB	46.0	45.2	48.5	53.8	63.9	69.4	72.4	71.3	65.1	58.5	53.4	49.6	(21)	
		MCWB	43.8	41.5	42.4	46.5	56.7	60.3	65.3	64.8	58.8	55.2	51.1	47.6	(22)	
	5%	DB	44.7	43.3	45.3	51.2	61.2	66.5	70.7	69.3	63.6	57.3	51.6	48.2	(23)	
		MCWB	42.8	40.8	40.5	45.8	54.6	58.6	64.3	63.1	58.9	54.1	49.3	46.3	(24)	
	10%	DB	43.2	41.8	43.2	49.1	58.4	64.3	68.8	67.5	62.5	56.1	50.2	46.7	(25)	
		MCWB	41.3	39.6	40.1	44.7	52.8	57.9	63.0	61.7	58.7	53.2	48.0	44.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.0	44.2	46.9	50.9	63.8	65.4	69.0	67.5	62.9	58.2	53.6	50.0	(27)	
		MCDB	48.1	45.9	53.4	54.7	67.7	69.1	73.4	71.2	64.1	59.0	54.6	50.8	(28)	
	2%	WB	44.7	42.8	43.3	48.7	58.3	62.1	67.3	65.9	61.8	56.3	51.6	48.3	(29)	
		MCDB	45.7	44.0	46.3	51.7	62.8	66.4	71.0	69.9	63.1	57.7	53.1	49.4	(30)	
	5%	WB	43.1	41.4	42.0	47.1	55.9	60.5	65.5	64.6	60.7	55.1	49.9	46.9	(31)	
		MCDB	44.3	43.1	44.2	50.1	59.7	64.6	69.4	68.4	62.6	56.5	51.4	48.1	(32)	
	10%	WB	41.6	39.7	40.8	45.2	53.8	59.1	63.9	63.2	59.4	53.8	48.3	45.2	(33)	
		MCDB	43.1	41.3	42.8	48.0	57.4	63.1	67.7	66.6	61.9	55.4	49.9	46.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.5	7.6	7.6	7.9	8.3	7.9	7.9	7.8	7.1	6.7	6.8	(35)	
		MCDBR	7.4	7.9	10.9	11.1	11.0	11.9	10.3	9.7	8.7	7.4	7.5	6.8	(36)	
	5% WB	MCWBR	7.2	6.2	7.1	7.3	7.1	7.6	6.6	6.5	6.7	6.5	7.8	7.0	(37)	
		MCWBR	6.8	7.0	8.8	9.0	9.2	8.6	8.3	7.7	7.0	6.3	7.2	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.325	0.343	0.382	0.373	0.358	0.378	0.379	0.354	0.338	0.308	0.273	(40)	
		taud	2.239	2.317	2.362	2.317	2.415	2.481	2.441	2.436	2.495	2.440	2.297	2.184	(41)	
	Ebn at Noon		169	222	254	261	273	279	270	261	250	217	163	136	(42)	
		Edh at Noon	15	22	28	34	33	31	32	30	24	20	14	11	(43)	
(44) All-Sky Solar Radiation	RadAvg		106	284	736	1237	1674	1905	1741	1339	864	394	143	75	(44)	
	RadStd		24	60	66	170	185	136	167	83	100	61	36	10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-259	N/A	N/A	

Nomenclature: See separate page